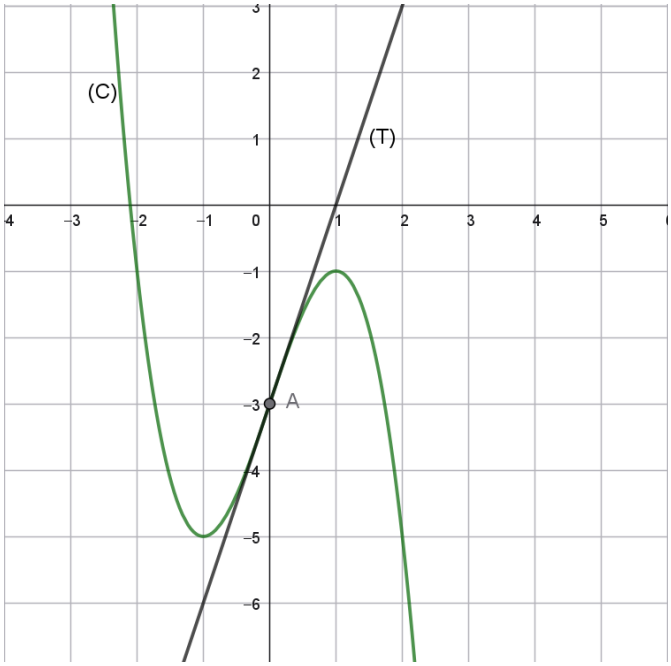
**Class: Grade 12 LS** **Subject: Mathematics**

**Teacher: Muhieddeen Iskandarani** **Worksheet: inverse function**

## Exercise 1:

In the adjacent figure:

- (C) is that of a function $f$ defined on $\mathbb{R}$.
- $(T)$is the tangent to (C) at the point A $(0, -3)$.




1) **Determine graphically:** $\lim\limits_{x\to-\infty} f(x)$ and $\lim\limits_{x\to+\infty} f(x)$.
2) Write the equation of the line $(T)$.
3) **Determine graphically:**
   a) $f(0), f(1), f(-1)$
   b) $f'(1), f'(-1), f'(0)$. Justify.
4) a) Show that $f$ admits an inverse function $f^{-1}$ for $x \in ]-1, 1[$.

   b) Find the domain of $f^{-1}$.

   c) Calculate $(f^{-1})'(-3)$.

   d) Find the equation of $(T')$, the tangent to $C^{-1}$

   (the curve of $f^{-1}$) at the point $(-3, 0)$.

5) Draw the table of variations of $f$.
6) Suppose that $f$ is defined by: $f(x) = ax^3 + bx^2 + 3x + c$ where $a, b, c$ are real constants.

   a) Determine the expression of $f'(x)$ ib terms of a and b.

   b) Show that $a = -1$, $b = 0$, $c = -3$.

7) Without calculating $\alpha$, prove that the equation $f(x) = 0$ admits a unique solution $\alpha$.

   Verify that $-2.2 < \alpha < -2.1$

8) Study according to the values of x the sign of $f(x)$.

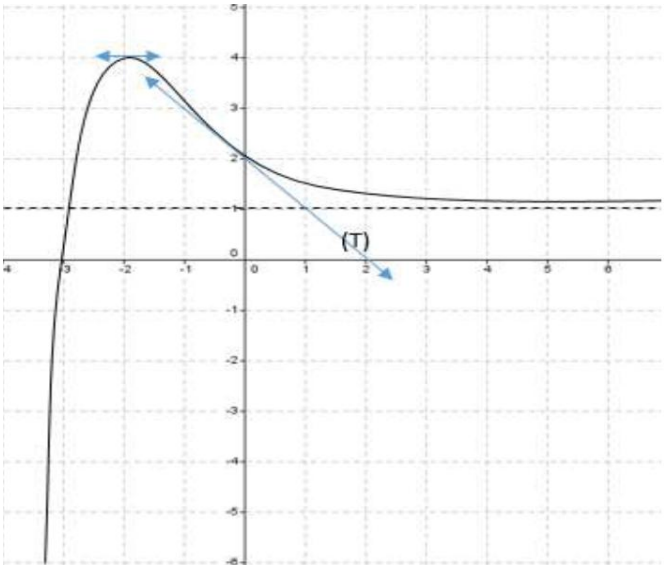
## Exercise 2:

The adjacent curve represents a function $f$ defined over $\mathbb{R}$,

in an orthonormal system.

The line $(T)$is tangent to the curve at the point $(0, 2)$

and passes through the point $(2, 0)$.




1) Determine graphically: $\lim\limits_{x\to+\infty} f(x)$ and $\lim\limits_{x\to-\infty} f(x)$.

2) Find graphically:$f'(-2)$ and $f'(0)$.Justify your answer.

3) Draw the table of variation of f.

4) a) Show that $f$admits over $[-2,+\infty[$ an inverse function $f^{-1}$.

b) Determine the domain of definition of $f^{-1}$.

c) Calculate $(f^{-1})'(2)$.

d) Find the equation of the tangent to the curve of $f^{-1}$at the point $(2,0)$. What do you notice? Justify.

e) Copy the curve of $f$over $[-2+\infty[$;and draw the curve of the inverse $f^{-1}$, showing the line $(T)$.

**Exercise 3:**

Let f, be the function defined over $]-\infty,-1[\cup]-1,+\infty[$ by $f(x)=\frac{x^2-x-1}{x-1}$ and designate by (C) its representative curve an orthonormal system.

1) Find the limits of $f$at the open boundaries of its domain. Deduce the equation of an asymptote $(C)$.

2) Show that $f$can be written in the form of $f(x)=ax+b+\frac{c}{x-1}$ where $a,b\ and\ c$ are real numbers to be determined.

3) a) Show that the line $(d)$of equation $y=x$ is an asymptote of $(C)$.

b) Study according to the values of x the relative position of $(C)$and $(d)$.

4) Calculate $f'(x)$and set up the table of variations of f.

5) Show that there is no point of $(C)$at which the tangent is parallel to $(d)$.

6) Draw $(C)$and its asymptotes.

7) Prove by three ways that the equation $f(x)=5$admits over $]1,+\infty[$ a unique root $\alpha$.

8) a) Show that f admits over $]1,+\infty[$ an inverse function $f^{-1}$

b) Find the domain of definition of $f^{-1}$.

c) Draw the curve of $f^{-1}$.

d) Calculate $(f^{-1})'(4)$.

e) Find the expression of $f^{-1}(x)$.